

THAARU Sepsis AI — Temporal Sequence Report

Sequence Generation Summary • Generated: July 19, 2026

1. Executive Summary

This phase generated **5,009,250** temporal sequences across **5** experiment configurations. Each sequence is a fixed-length window of standardized clinical features, paired with a binary sepsis label aligned to the specified prediction horizon.

2. Experiment Dataset Summary

Config	Window	Horizon	Train Seqs	Val Seqs	Test Seqs	Train +Rate
w6_h0	6h	+0h	945,261	201,222	204,047	1.87%
w12_h0	12h	+0h	776,999	165,213	167,979	1.91%
w24_h0	24h	+0h	472,453	100,373	102,520	2.48%
w12_h3	12h	+3h	695,059	147,719	150,382	2.00%
w12_h6	12h	+6h	615,848	130,823	133,352	2.13%

3. Validation Results

w6_h0: ■ ALL PASSED

w12_h0: ■ ALL PASSED

w24_h0: ■ ALL PASSED

w12_h3: ■ ALL PASSED

w12_h6: ■ ALL PASSED

4. Per-Configuration Details

w6_h0 — Minimal observation: can 6h detect sepsis?

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	945,261	17,643	927,618	1.87%	52.6:1
Validation	201,222	3,690	197,532	1.83%	53.5:1
Test	204,047	3,803	200,244	1.86%	52.6:1

w12_h0 — Standard 12h observation window

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	776,999	14,827	762,172	1.91%	51.4:1
Validation	165,213	3,061	162,152	1.85%	53.0:1
Test	167,979	3,221	164,758	1.92%	51.1:1

w24_h0 — Extended observation: does more history help?

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	472,453	11,715	460,738	2.48%	39.3:1
Validation	100,373	2,241	98,132	2.23%	43.8:1
Test	102,520	2,576	99,944	2.51%	38.8:1

w12_h3 — Early warning: 3-hour advance prediction

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	695,059	13,887	681,172	2.00%	49.0:1
Validation	147,719	2,821	144,898	1.91%	51.4:1
Test	150,382	3,017	147,365	2.01%	48.8:1

w12_h6 — Early warning: 6-hour advance prediction

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	615,848	13,089	602,759	2.13%	46.0:1
Validation	130,823	2,585	128,238	1.98%	49.6:1
Test	133,352	2,841	130,511	2.13%	45.9:1